FACE AND EYE DETECTION USING OPENCV

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A Real-Time / Field-Based Research Project (22AM284) report submitted to the Jawaharlal Nehru Technological University, Hyderabad

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1.1 PROBLEM STATEMENT

The " Real-Time Face and Eye Detection Using Open CV" project aims to address the need for an efficient and accurate system that can detect faces and eyes in live video streams. Current face and eye detection technologies often struggle with real-time performance, accuracy in varied lighting conditions, and handling different orientations and facial expressions. These challenges hinder their effectiveness in critical applications such as security surveillance, biometric authentication, and interactive systems.

The primary problem this project seeks to solve is the development of a robust, real-time detection system capable of identifying faces and eyes with high accuracy and minimal latency. Leveraging OpenCV's pre-trained Haar cascade classifiers, the project will create a solution that processes video frames from a webcam, converting them to grayscale for more effective detection. The system must balance speed and accuracy, ensuring it can operate efficiently in real-time while maintaining reliable detection under diverse conditions.

This project aims to provide a user-friendly interface that highlights detected faces and eyes in the video stream, offering immediate visual feedback. By solving these issues, the project will enhance the applicability of face and eye detection systems in various fields, including security, user interaction, and healthcare monitoring.